The Difference between Char, Nchar, Varchar and Nvarchar

To compare these four types effectively, let's summarize their behaviours in a table structure:

Property	CHAR	NCHAR	VARCHAR	NVARCHAR
Description	Fixed-length non-Unicode character	Fixed-length Unicode character	Variable-length non-Unicode character	Variable-length Unicode character
Storage	1 byte per character	2 bytes per character	1 byte per character	2 bytes per character
Allocation	Allocates full length	Allocates full length	Allocates only needed	Allocates only needed
Unicode Support	No	Yes	No	Yes
Example Practical Usage	State or Country Codes	Unicode Country Code	Product Description	Customer feedback

The differences between **CHAR**, **NCHAR**, **VARCHAR**, and **NVARCHAR** depend on two main factors: how they store data (**fixed vs. variable length**) and what type of characters they support (**non-Unicode vs. Unicode**).

Summary Comparison Table

Data Type	Length	Encoding	Bytes per Char	Best Use Case
CHAR	Fixed	Non-Unicode (ASCII)	1 byte	Codes of exact length (e.g., ISO country codes like 'US')
NCHAR	Fixed	Unicode (UTF-16)	2 bytes	Fixed-length data with international characters
VARCHAR	Variable	Non-Unicode (ASCII)	1 byte	Varying length data in a single language (e.g., emails)
NVARCHAR	Variable	Unicode (UTF-16)	2 bytes	Varying length multilingual data (e.g., names, addresses)

Key Distinctions

- **Fixed vs. Variable Length**
    - **Fixed (CHAR / NCHAR)**: Always uses the full defined size. If the data is shorter than the limit, the database pads the remaining space with blanks. This can offer slightly better performance for consistently sized fields but wastes space for others.
    - **Variable (VARCHAR / NVARCHAR)**: Only uses the space required for the actual data plus a small overhead (typically 2 bytes) to store the length. This is more storage-efficient for data that varies in length.
- **Non-Unicode vs. Unicode ("N")**
    - **Non-Unicode (CHAR / VARCHAR)**: Designed for standard ASCII characters. They typically use **1 byte** per character and are limited to a single code page (one language set).

    - **Unicode (NCHAR / NVARCHAR)**: The "N" stands for "National" character set. These use **2 bytes** per character (UCS-2/UTF-16), allowing them to store characters from almost any language, including Chinese, Arabic, and Japanese.

- **Storage Limits**
    - **Non-Unicode**: Max length is typically 8,000 characters.
    - **Unicode**: Max length is typically 4,000 characters (because each character takes double the space).

Note: VARCHAR(MAX) and NVARCHAR(MAX) can store up to 2GB of data.

Comparison with Unicode

Feature	**Non-Unicode**	**Unicode**
Storage	Typically 1 byte per character	2 or 4 bytes per character (UTF-8, UTF-16)
Language Support	Limited (single script/language)	Global (all languages & symbols)
Compatibility	Often causes issues across systems	High consistency across platforms
Database Type	char, varchar, text	nchar, nvarchar, ntext